

AETHEXER / EXITZ

Universal Truth Layer & Carbon Displacement Token

Prepared by ZerO | April 2026

What Is Aethexer?

Aethexer is the **Universal Truth Layer** — a proprietary protocol that provides military-grade cryptographic security for digital transactions while simultaneously generating measurable environmental benefit. It is the intellectual property of ZerO and serves as the cryptographic backbone of the Bedrock ESG platform.

The core insight behind Aethexer is simple but powerful: every time a digital process replaces a physical one — every armored truck that doesn't drive, every courier package that becomes a secure digital transfer — there is a measurable reduction in carbon emissions. Aethexer captures, verifies, and tokenizes that reduction.

The result is **EXITZ** — a carbon displacement token that represents the environmental benefit of choosing digital over physical.

The Problem

The modern economy still moves enormous value through physical infrastructure — armored vehicles, courier services, secure transport. This physical movement generates significant carbon emissions that are rarely accounted for, never offset, and completely avoidable when digital alternatives exist.

At the same time, digital security protocols exist in isolation from environmental accounting. A bank encrypting a wire transfer doesn't receive credit for the armored truck it didn't dispatch. A law firm sending documents through a secure portal doesn't quantify the courier emissions it avoided.

Aethexer bridges this gap by wrapping every digital transaction in military-grade security while simultaneously calculating and recording the carbon displacement it creates.

How It Works

Armored Digital Packets

The fundamental unit of the Aethex Protocol is the **Armored Digital Packet** — a cryptographically secured data container that replaces the need for physical secure transport.

Think of it as the digital equivalent of an armored truck: it carries valuable cargo (data, documents, financial instructions) through a hostile environment (the internet) with three layers of protection that would make any physical security system envious.

Each Armored Digital Packet carries with it a calculated carbon displacement — the measurable environmental benefit of this digital transfer existing instead of a physical one.

Three-Tier Security Model

Every Armored Digital Packet is protected by three independent security layers, each addressing a different threat vector:

Tier 1 — Ballistic Layer

The innermost layer provides AES-GCM-256 encryption — the same encryption standard used by intelligence agencies and military systems worldwide. Every piece of data inside the packet is individually encrypted with a unique key and initialization vector, making interception meaningless without the decryption credentials.

This layer answers the question: “If someone intercepts this packet, can they read it?” The answer is no.

Tier 2 — Guardian Layer

The middle layer implements 2-of-2 Multi-Signature Verification. Both the sender and the receiver must cryptographically sign the packet for it to be valid. This prevents unauthorized access even if encryption keys are compromised — without both signatures, the packet cannot be opened.

This layer answers the question: “If someone has the key, can they access this without authorization?” The answer is no — both parties must consent.

Tier 3 — Stealth Layer

The outermost layer uses Zero-Knowledge Proofs (ZKP) — a cryptographic technique that allows one party to prove they possess certain information without revealing the information itself. This means the packet’s integrity and authenticity can be verified by anyone, without exposing its contents.

This layer answers the question: “Can we verify this packet is legitimate without seeing what’s inside?” The answer is yes.

Packet Lifecycle

Every Armored Digital Packet moves through a defined lifecycle:

1. **Created** — Payload is encrypted with AES-GCM-256 (Ballistic layer activated)
2. **Signed** — Both parties apply their cryptographic signatures (Guardian layer activated)
3. **Verified** — Zero-Knowledge Proof is generated (Stealth layer activated)
4. **In Transit** — Packet moves between parties through any network
5. **Delivered** — Received and verified at destination
6. **Settled** — Carbon displacement is calculated and recorded on the blockchain

At each stage, the packet maintains full cryptographic integrity. Any tampering at any point invalidates the entire chain of proofs.

Carbon Displacement — The Environmental Engine

The Aethex Protocol doesn’t just secure data — it quantifies the environmental benefit of doing so.

The Displacement Factor

Physical armored transport — the trucks, security vehicles, and personnel that move valuable cargo — generates approximately **2.7 kg of CO₂e per mile** when accounting for:

- Vehicle fuel consumption (armored vehicles are significantly heavier than standard transport)
- Security escort overhead (additional vehicles, personnel transport)
- Route inefficiency (secure routes are often longer than optimal routes)
- Idle time and staging (engines running during loading, unloading, verification)

When an Armored Digital Packet replaces a physical transport event, the distance that would have been driven is multiplied by this 2.7 kg CO₂e/mile displacement factor to calculate the verified carbon savings.

Current Performance

As of the most recent data:

Metric	Value
Total Packets Processed	779
Total CO ₂ e Displaced	~6.7 tonnes
Average Displacement per Packet	~8.6 kg CO ₂ e
Settlement Status	Active

These are real, calculated displacement values based on the digital-versus-physical comparison methodology.

Settlement Economics

Every Armored Digital Packet carries a settlement fee of 0.5% of the transaction value, with a minimum packet value of \$100 USD. Batch transfers are supported up to 1,000 packets per operation, enabling high-throughput processing for enterprise clients.

The EXITZ Token

EXITZ is the native carbon displacement token of the Aethexer ecosystem. Each token represents one verified unit of carbon displacement — the environmental benefit created when a digital process replaces a physical one.

Token Design Philosophy

EXITZ tokens are designed to be **precise and indivisible**. Each token represents a discrete displacement event with zero decimal places — you either displaced carbon or you didn't. There are no fractional EXITZ tokens, which prevents the kind of artificial precision that undermines credibility in conventional carbon markets.

Token Specifications

Property	Detail
Standard	Solana Token-2022 (SPL)
Network	Solana
Decimals	0 — whole units only
Mint Address	44SrHT9Qwyz2jiiJkRF1zHKF2NFTgTC8m6ubPT-b5qJBJ
Mint Authority	Aethexer Sentinel — sole authority to issue new tokens
Freeze Authority	Aethexer Sentinel — can freeze tokens if fraud is detected

On-Chain Metadata

Every EXITZ token carries on-chain metadata that links it permanently to the Bedrock ESG verification infrastructure:

Field	Value	Purpose
Name	EXITZ	Token identity
Symbol	EXITZ	Trading symbol
Backtrace ID	BT-C9C4C5	Links to Bedrock ESG audit trail
Engine	Bedrock ESG	Identifies the verification platform
Protocol	Aethex Universal Truth Layer	Identifies the security protocol
Standard	ISO 14064-2	GHG quantification standard compliance

This metadata is written directly to the blockchain at token creation and cannot be modified after the fact. Any EXITZ token can be traced back through the Bedrock ESG platform to the specific environmental event that generated it.

Genesis Emission

The first EXITZ token — the Genesis Emission of 1.0 EXITZ — has been minted and confirmed on the Solana blockchain. This initial token serves as proof of the minting infrastructure and establishes the token's on-chain existence.

Retroactive Harvest

One of Aethexer’s most powerful capabilities is the ability to look backward — to analyze historical enterprise data and retroactively calculate the environmental impact that was never previously measured.

What It Does

The Retroactive Harvest system ingests data from enterprise resource planning (ERP) systems and audits it against current environmental standards. Organizations running SAP, Oracle, or Viewpoint (common in construction) can feed their historical procurement, logistics, and materials data into the system.

The Harvest engine then:

1. **Ingests** the ERP data across procurement, logistics, and materials categories
2. **Audits** every record against 2026 Caltrans Environmental Standards and ISO 14064-2 requirements
3. **Calculates** the retroactive environmental impact — what was the carbon footprint of these historical operations?
4. **Mints Material Passports** — digital documents that travel with physical materials, containing full chain-of-custody, environmental impact data, and compliance status
5. **Attests** everything on the Solana blockchain with the Aethexer Sentinel

Supported ERP Systems

System	Coverage
SAP (S/4HANA)	Full procurement, logistics, and materials modules
Oracle (Cloud ERP)	Enterprise resource and supply chain data
Viewpoint (Vista / Spectrum)	Construction-specific project management and materials tracking

Material Passports

A Material Passport is a digital twin of a physical material’s journey through the supply chain. It contains:

- **Chain of Custody** — every hand the material passed through, with timestamps
- **Environmental Impact** — embodied carbon, transport emissions, processing energy
- **Compliance Status** — current standing against applicable environmental standards
- **Blockchain Attestation** — Solana proof linking the passport to the Bedrock ESG audit trail

Material Passports are increasingly required by regulators and procurement departments that need to verify the environmental footprint of their supply chain. The Retroactive Harvest system can generate these passports from historical data, bringing organizations into compliance without requiring them to have tracked this information in real time.

The Aethexer Sentinel

All operations across both Aethexer and Bedrock ESG flow through a single institutional cryptographic identity: the **Aethexer Sentinel** (designated **Sentinel Node-01**).

This is a Solana wallet address — **9tsq8qB4P9uPRSHrjXLaDQJd93nuo4wEEtkR1tAJRAQ3** — that serves as the sole signing authority for:

- All blockchain attestations
- EXITZ token minting
- Material Passport issuance
- Settlement transaction recording
- Audit proof generation

The Sentinel designation reflects the institutional nature of this authority — it is not a personal key but a sovereign platform identity. Every operation on the platform can be independently verified by checking that it was signed by the Aethexer Sentinel. This creates a simple, powerful trust model: if the Sentinel signed it, the Bedrock ESG platform verified it.

The backtrace identifier **BT-C9C4C5** accompanies every Sentinel-signed operation, providing a human-readable audit trail reference that links on-chain transactions back to the platform's internal records.

How Aethexer and Bedrock ESG Connect

Aethexer is not a separate product — it is the cryptographic and environmental accounting layer that powers Bedrock ESG. They share:

Shared Resource	Why It Matters
Database	Single source of truth for all environmental and transaction data
Blockchain Identity	Same Aethexer Sentinel signs everything — unified provenance
Authentication	Same user accounts and role-based access control
Compliance Infrastructure	Same ISO standards, same emission factor databases

The data flow works like this:

1. Bedrock ESG modules generate verified environmental data (carbon calculations, biochar CDR, metal recovery)
2. The Universal Ledger records these as audited entries
3. The Aethex Settlement Engine processes these entries and calculates carbon displacement
4. EXITZ tokens are minted to represent the displacement

5. Everything is attested on-chain through Aethexer Sentinel

This integration means that the environmental benefit of every asphalt calculation, every biochar batch, and every metal recovery log ultimately flows through the Aethex Protocol and can be expressed as EXITZ tokens.

The aethexer.com Landing Page

The public-facing website at **aethexer.com** serves as the information and brand identity portal for the Aethex Protocol. It was built as a separate project to maintain clean architectural boundaries between the marketing presence and the operational infrastructure.

The landing page presents the protocol's capabilities, security model, and value proposition to potential partners and users, while the actual protocol engine runs on the Bedrock ESG infrastructure at bedrockesg.com.

This separation means that the brand can evolve independently of the technology, and the technology can be updated without affecting the public-facing marketing presence.

Why This Matters — For Partners and Investors

The Market Opportunity

The intersection of digital security and environmental accounting is largely unoccupied. Cybersecurity companies don't think about carbon displacement. Carbon credit platforms don't think about cryptographic security. Aethexer operates in both spaces simultaneously, creating a category that doesn't yet have established competition.

The Moat

- **Proprietary protocol** — the three-tier Ballistic/Guardian/Stealth architecture is ZerO's intellectual property
- **On-chain proof** — every claim is independently verifiable on public blockchains
- **Institutional databases** — calculations are grounded in the same emission factor databases used by regulators and academic institutions
- **ERP integration** — the Retroactive Harvest system creates immediate value from existing enterprise data
- **Token economics** — EXITZ tokens create a liquid, tradeable representation of verified environmental impact

The Vision

Every digital transaction that replaces a physical process is an environmental event. Aethexer makes that event measurable, verifiable, and valuable. As more of the economy moves digital, the displacement opportunity grows — and Aethexer is positioned to capture and tokenize that displacement at scale.

Key Terms

Term	What It Means
Aethexer	The protocol system — the “Universal Truth Layer”
EXITZ	The carbon displacement token minted when digital replaces physical
Armored Digital Packet	A three-layer encrypted data container that replaces physical transport
Aethexer Sentinel (Node-01)	The institutional blockchain wallet that signs all platform operations (formerly Kato Service Identity)
Backtrace (BT-C9C4C5)	The audit trail ID linking on-chain records to platform data
Displacement Factor	2.7 kg CO ₂ e per mile — the carbon cost of physical armored transport
Retroactive Harvest	System for mining historical ERP data for environmental impact
Material Passport	Digital twin documenting a material’s environmental journey
Ballistic Layer	Military-grade AES-GCM-256 encryption (Tier 1)
Guardian Layer	2-of-2 multi-signature authorization (Tier 2)
Stealth Layer	Zero-Knowledge Proof verification (Tier 3)
Zero	The organization behind both Bedrock ESG and Aethexer
Bedrock ESG	The sovereign environmental verification platform that hosts the Aethex engine
Universal Ledger	The master accounting system inside Bedrock ESG

Companion Document

This brief is designed to be read alongside the **Bedrock ESG Executive Brief**, which covers the full verification platform including the Asphalt LCA Calculator, Biochar CDR Engine, Metal Recovery Tracker, Digital Vault, Universal Ledger, and Business Submission Portal.

Together, these two documents provide a complete picture of the ZerO ecosystem — from raw environmental data to verified, tokenized, tradeable carbon displacement credits.

Contact & Access

- **Protocol Engine:** bedrockesg.com (operational infrastructure)
 - **Landing Page:** aethexer.com (public information portal)
 - **Organization:** ZerO
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This document is prepared for partner and investor review. It describes the capabilities and architecture of the Aethex Protocol and EXITZ token as of April 2026. No credentials, private keys, or internal access information are included.